

Explicit Bounds for Some Functions of Prime Numbers

Author(s): Barkley Rosser

Source: *American Journal of Mathematics*, Jan., 1941, Vol. 63, No. 1 (Jan., 1941), pp. 211-232

Published by: The Johns Hopkins University Press

Stable URL: https://www.jstor.org/stable/2371291

REFERENCES
Linked references are available on JSTOR for this article:
https://www.jstor.org/stable/2371291?seq=1&cid=pdf-reference#references_tab_contents
You may need to log in to JSTOR to access the linked references.

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at https://about.jstor.org/terms



The Johns Hopkins University Press is collaborating with JSTOR to digitize, preserve and extend access to *American Journal of Mathematics*

This content downloaded from
154.59.125.115 on Sun, 01 Nov 2026 12:04:59 UTC
All use subject to https://about.jstor.org/terms

EXPLICIT BOUNDS FOR SOME FUNCTIONS OF PRIME NUMBERS.*

By BARKLEY ROSSER.

Summary of results. Counting 2 as the first prime, we denote by $\pi(x)$, $p(n)$, and $\theta(x)$, respectively, the number of primes less than or equal to x, the n-th prime, and the logarithm of the product of all primes less than or equal to x. It is known that for each positive constant A, there is a constant N for which the three following statements hold true.

If $N \leq x$, then

$$\frac{x}{\log x - 1 + A} < \pi(x) < \frac{x}{\log x - 1 - A}.$$

If $N \leq n$, then

$$n \log n + n \log \log n - n - An < p(n) < n \log n + n \log \log n - n + An.$$

If $N \leq n$, then

$$\left(1 - \frac{A}{\log x}\right) x < \theta(x) < \left(1 + \frac{A}{\log x}\right) x.$$

Moreover, these three statements are essentially interdeducible, in the sense that if one of them can be proved for a certain value, A^*, of A, then the other two can be proved with the value $A^* + \epsilon$ for A, where ϵ depends on N.

Heretofore the question of determining the N which goes with a particular A has received no attention. Moreover, the question of how small A can be taken without requiring that N become large has been neglected.

Theorem 22 of this paper furnishes an explicit answer (not the best possible) to the first question. For the second question the answer $A = 3$ is given. Again this is not the best possible, and a partial proof is given that we can take $A = 1$. In particular, for $A = 3$, we have:

THEOREM 29. *If* $55 \leq x$, *then:*

A. $\dfrac{x}{\log x + 2} < \pi(x) < \dfrac{x}{\log x - 4}.$

B. $x \log x + x \log \log x - 4x < p([x]) < x \log x + x \log \log x + 2x.$

C. $\left(1 - \dfrac{3}{\log x}\right) x < \theta(x) < \left(1 + \dfrac{3}{\log x}\right) x.$

* Received June 28, 1940.

This content downloaded from
154.59.125.115 on Sun, 01 Nov 2026 12:04:59 UTC
All use subject to https://about.jstor.org/terms

For $A = 1$, we have the partial result embodied in the following seven theorems.

THEOREM 30. *If* $e^{2000} \leq x$, *then:*

A. $\dfrac{x}{\log x} < \pi(x) < \dfrac{x}{\log x - 2}$.

B. $x \log x + x \log\log x - 2x < p([x]) < x \log x + x \log\log x$.

C. $\left(1 - \dfrac{1}{\log x}\right) x < \theta(x) < \left(1 + \dfrac{1}{\log x}\right) x$.

THEOREM 26. *If* $17 \leq x \leq e^{100}$, *then* $\dfrac{x}{\log x} < \pi(x)$.

THEOREM 25. *If* $e^2 < x \leq e^{100}$, *then* $\pi(x) < \dfrac{x}{\log x - 2}$.

THEOREM 27. *If* $1 < n \leq e^{100}$, *then* $n \log n + n \log\log n - 2n < p(n)$.

THEOREM 28. *If* $6 \leq n \leq e^{95}$, *then* $p(n) < n \log n + n \log\log n$.

THEOREM 23. *If* $41 \leq x \leq e^{100}$, *then* $\left(1 - \dfrac{1}{\log x}\right) x < \theta(x)$.

THEOREM 24. *If* $1 < x \leq e^{100}$, *then* $\theta(x) < \left(1 + \dfrac{1}{\log x}\right) x$.

For $0 < x \leq 1{,}000{,}000$, these results were derived from the very sharp theorem:

If $0 < x \leq 1{,}000{,}000$, *then* $x - 2.78x^{\frac{3}{4}} < \theta(x) < x$;

which was proved essentially by comparison with Lehmer's "List of Prime Numbers." For $1{,}000{,}000 \leq x$, analytical methods were used. It is remarkable that in the ranges $1{,}000{,}000 \leq x \leq e^{95}$ and $e^{2000} \leq x$, the analytical methods enable one to take $A = 1$, whereas in the intermediary range $e^{95} \leq x \leq e^{2000}$ one has to be satisfied with a larger value of A. This peculiar situation is apparently due to the insufficiency of our present knowledge about the zeros of the Riemann zeta function. A significant increase in our present knowledge would undoubtedly enable one to take $A = 1$ in the intermediary range also. For instance, if $\beta + i\gamma$ is a typical complex zero of the zeta function, it would suffice to know that

$$\beta < 1 - \frac{1}{6 \log \gamma}$$

for $1400 \leq \gamma$.

This content downloaded from
154.59.125.115 on Sun, 01 Nov 2026 12:04:59 UTC
All use subject to https://about.jstor.org/terms

References to the bibliography will consist of a number referring to the numbered bibliography at the end of the paper followed by the necessary page references.

Tables of $\psi(x)$, $\theta(x)$, and $p(n)$. We define

$$\psi(x) = \sum_{p^m \leq x} \log p.$$

The following relations connect $\theta(x)$ and $\psi(x)$:

$$(1) \qquad \psi(x) = \sum_{n=1}^{[x]} \theta(x^{1/n}) = \theta(x) + \theta(x^{\frac{1}{2}}) + \theta(x^{1/3}) + \cdots$$

$$\theta(x) = \sum_{n=1}^{[x]} \mu(n)\psi(x^{1/n})$$

$$(2) \qquad = \psi(x) - \psi(x^{1/2}) - \psi(x^{1/3}) - \psi(x^{1/5}) + \psi(x^{1/6}) - \cdots.$$

Gram has given a table of $\psi(x)$ to eight decimal places for $1 \leq x \leq 2000$ (see 1, pp. 281-288). As a check on Gram's table, the present author computed $\psi(x)$ for $1 \leq x \leq 2000$ by use of a six place table of natural logarithms and found no discrepancies outside the limits of accuracy of the computation. By comparing the first few entries of Gram's table with values computed from eight place, nine place, ten place, and eleven place values of the natural logarithms of the primes, it was apparent that Gram must have used eight place values in computing his table. Hence the eighth place in Gram's table is not reliable.

By use of (2) and Gram's table, one can readily compute $\theta(x)$ for $1 \leq x \leq 2000$. In order to facilitate the computation of $\theta(x)$ for $x \leq 10{,}000$, Table I was prepared (see end of paper). Table I is auxiliary to a table of Jones (2, pp. 114-117. Note two errors: $\log 6899 = 8.839132$ and $\log 7853 = 8.968651$). Jones's table gives the natural logarithms to six decimal places of the primes from 2000 to 10,000. To compute $\theta(x)$, one can take the nearest entry below x in Table I, and add the logarithms of the intervening primes as given in Jones's table. As a check, one can take the next entry above x, and subtract the logarithms of the intervening primes. So that this check will come out exact, Table I was computed from Jones's table, and all six places were retained, though the last is quite unreliable. Table I was checked by adding up seven place common logarithms of the primes, and multiplying the sums by $\log_e 10$.

Lehmer's List of Prime Numbers (3, pp. 1-135) is a tabulation of $p(n-1)$ as a function of n for $1 \leq n \leq 675{,}000$ (Lehmer takes 1 as the first prime). From this table, one can read off $\pi(x) + 1$ for $1 \leq x \leq 10{,}006{,}721$.

Relations between $\pi(x)$, $p(n)$, and $\theta(x)$. Obvious relations are

This content downloaded from
154.59.125.115 on Sun, 01 Nov 2026 12:04:59 UTC
All use subject to https://about.jstor.org/terms

$$\pi(p(n)) = n \tag{3}$$

$$p(\pi(x)) \leq x \tag{4}$$

$$\theta(p(n)) = \sum_{r=1}^{n} \log p(r). \tag{5}$$

By use of a theorem connecting sums with integrals (4, Theorem A, p. 18), we get

$$\theta(x) = \pi(x) \log x - \int_2^x \frac{\pi(y)\,dy}{y} \tag{6}$$

$$\pi(x) = \frac{\theta(x)}{\log x} + \int_2^x \frac{\theta(y)\,dy}{y \log^2 y}. \tag{7}$$

From these we now proceed to deduce relations which hold between functions which bound $\pi(x)$, $p(n)$, and $\theta(x)$. In constructing bounding functions for $\pi(x)$, the logarithmic integral,

$$li(x) = \lim_{\eta \to +0} \left\{ \int_0^{1-\eta} \frac{dy}{\log y} + \int_{1+\eta}^x \frac{dy}{\log y} \right\},$$

is very convenient to use, so we call attention to a few tables of $li(x)$ and the related function

$$Ei(x) = li(e^x).$$

The most complete table of $Ei(x)$ is Table VII in Vol. 1 of the Mathematical Tables of the British Association for the Advancement of Science. A short table of $Ei(x)$ is given in Chapter VI of Funktionentafeln, by Jahnke and Emde. Lehmer gives a table of $li(x)$ correct to the nearest integer (3, pp. xiii-xvi. The column headed "Tchebycheff" contains vaules of $li(x)$, and not values of $\int_2^x \frac{dy}{\log y}$ as Lehmer claims).

LEMMA 1. $\int_2^{e^{2.4}} \frac{\theta(y)\,dy}{y \log^2 y} < li(e^{2.4}) - \frac{e^{2.4}}{2.4}$.

Proof. The expression on the right has the value 2.008, whereas, by (7), the expression on the left has the value 1.773.

A similar proof using (6) holds for the next two lemmas.

LEMMA 2. $\int_2^{e^4} \frac{\pi(y)\,dy}{y} > 4li(e^4) - e^4 + 2e^2 - 4li(e^2)$.

LEMMA 3. $\int_2^{e^3} \frac{\pi(y)\,dy}{y} < 3li(e^3) - e^3$.

This content downloaded from
154.59.125.115 on Sun, 01 Nov 2026 12:04:59 UTC
All use subject to https://about.jstor.org/terms

LEMMA 4. *If* $\theta(x) < x$ *for* $e^{2.4} \leq x \leq K$, *then* $\pi(x) < li(x)$ *for* $e^{2.4} \leq x \leq K$.

Proof. Under the hypothesis of the theorem we can deduce from (7) and Lemma 1 that, if $e^{2.4} \leq x \leq K$, then

$$\pi(x) < \frac{x}{\log x} + li(e^{2.4}) - \frac{e^{2.4}}{2.4} + \int_{e^{2.4}}^{x} \frac{y\,dy}{y \log^2 y}.$$

However, integration by parts gives

$$\int \frac{y\,dy}{y \log^2 y} = -\frac{y}{\log y} + \int \frac{dy}{\log y},$$

from which the lemma follows.

For the next two lemmas, the proof is similar and makes use of the fact that integration by parts gives

$$\int \frac{li(y)\,dy}{y} = \log y\, li(y) - y,$$

and that if $y = z^2$

$$\int \frac{li(y^{\frac{1}{2}})\,dy}{y} = 2 \int \frac{li(z)\,dz}{z}.$$

LEMMA 5. *If* $li(x) - li(x^{\frac{1}{2}}) < \pi(x)$ *for* $e^4 \leq x \leq K$, *then*

$$\theta(x) < x - 2x^{\frac{1}{2}} + \log x(\pi(x) - li(x) + li(x^{\frac{1}{2}}))$$

for $e^4 \leq x \leq K$.

LEMMA 6. *If* $\pi(x) < li(x)$ *for* $e^3 \leq x \leq K$, *then*

$$x - \log x(li(x) - \pi(x)) < \theta(x)$$

for $e^3 \leq x \leq K$.

COROLLARY. *If* $li(x) - li(x^{\frac{1}{2}}) < \pi(x) < li(x)$ *for* $e^3 \leq x \leq K$, *then* $x - \log x\, li(x^{\frac{1}{2}}) < \theta(x)$ *for* $e^3 \leq x \leq K$.

LEMMA 7. *For* $e^4 \leq x$, $li(x) < \dfrac{x}{\log x - 2}$.

Proof. For $x = e^4$, the inequality is true, and for $e^4 \leq x$,

$$\frac{d}{dx} li(x) \leq \frac{d}{dx} \frac{x}{\log x - 2}.$$

LEMMA 8. *For* $e^4 \leq x$, $\dfrac{x}{\log x} < li(x) - li(x^{\frac{1}{2}})$.

Similar proof.

This content downloaded from
154.59.125.115 on Sun, 01 Nov 2026 12:04:59 UTC
All use subject to https://about.jstor.org/terms

LEMMA 9. *For* $n \geq 5$, $\theta(p(n)) > n \log n + n \log \log n - n - li(n)$.

Proof. Since $p(r) > r \log r$ (5, Theorem 1, p. 37), we have by (5),

$$\begin{aligned}\theta(p(n)) &\geq \theta(p(5)) + \sum_{r=6}^{n} \log(r \log r) \\ &\geq \theta(p(5)) + \int_5^n \log x\, dx + \int_5^n \log \log x\, dx \\ &\geq \theta(p(5)) + n \log n - n - 5 \log 5 + 5 \\ &\quad + n \log \log n - 5 \log \log 5 + li(5) - li(n) \\ &> n \log n + n \log \log n - n - li(n).\end{aligned}$$

LEMMA 10. *If* $16 \leq n$ *and if* $p(r) < r \log r + r \log \log r$ *for* $16 \leq r < n$, *then*

$$\theta(p(n)) < n \log n + n \log \log n - n + \frac{n \log \log n}{\log n}.$$

Proof. Since $p(n) < n \log n + 2n \log \log n$ for $n \geq 3$ (5, Theorem 2, p. 40), we have

$$\begin{aligned}\theta(p(n)) &< \theta(p(15)) + \sum_{r=16}^{n-1} \log(r \log r + r \log \log r) + \log(n \log n + 2n \log \log n) \\ &< \theta(p(15)) + \int_{16}^{n} \log x\, dx + \int_{16}^{n} \log(\log x + \log \log x)\, dx \\ &\quad + \log(n \log n + 2n \log \log n) \\ &< \theta(p(15)) + n \log n - n - 16 \log 16 + 16 + n \log(\log n + \log \log n) \\ &\quad - 16 \log(\log 16 + \log \log 16) - \int_{16}^{n} \frac{(\log x + 1)\, dx}{\log x (\log x + \log \log x)} \\ &\quad + \log(n \log n + 2n \log \log n).\end{aligned}$$

However

$$\log(\log n + \log \log n) = \log \log n + \log\left(1 + \frac{\log \log n}{\log n}\right) < \log \log n + \frac{\log \log n}{\log n},$$

and

$$\int_{16}^{n} \frac{(\log x + 1)\, dx}{\log x (\log x + \log \log x)} > \int_{16}^{n} \frac{dx}{2 \log x},$$

and

$$\begin{aligned}&\theta(p(15)) - 16 \log 16 + 16 - 16 \log(\log 16 + \log \log 16) \\ &\quad - \tfrac{1}{2}(li(n) - li(16)) + \log(n \log n + 2n \log \log n) < 0.\end{aligned}$$

The range $0 < x \leq 1{,}000{,}000$.

THEOREM 1. *For* $11 \leq x \leq 1{,}000{,}000$, $li(x) - li(x^{\frac{1}{2}}) < \pi(x)$.

Proof. The interval $0 < x \leq 1{,}000{,}000$ was divided into convenient sub-

This content downloaded from
154.59.125.115 on Sun, 01 Nov 2026 12:04:59 UTC
All use subject to https://about.jstor.org/terms

intervals. Corresponding to each subinterval, I_r, a linear function, $Ax + B$, was determined so that

$$li(x) - li(x^{\frac{1}{2}}) \leq Ax + B$$

for x in I_r. Then by comparison with Lehmer's List of Prime Numbers, it was determined that $Ax + B + 1 < \pi(x) + 1$ for x in I_r. To show how this can be done quickly, we exhibit a specimen of the computation. By getting the equation of the tangent to the curve

$$y = li(x) - li(x^{\frac{1}{2}})$$

at $x = e^{13.4}$, and by noting that the curve is convex upward, we ascertain that

$$li(x) - li(x^{\frac{1}{2}}) < 4{,}323.1 + \frac{x}{13.41651}$$

for $x > 2$. So we undertake to show that

$$4{,}324.1 + \frac{x}{13.41651} < \pi(x) + 1$$

in I_r. Put $x = e^{13.4} + 13.41651\, y = a + by$. Then it suffices to show that

$$53{,}518 + y \leq \pi(a + by) + 1 \tag{8}$$

for $a + by$ in I_r. Now, using a Monroe High Speed Adding Calculator (or any computing machine of similar construction), put 53,518 on the upper dials, put a (that is, 660,003.22477) on the lower dials, and b (that is, 13.41651) on the keyboard. Now if one holds the + bar down for y revolutions, $53{,}518 + y$ will appear on the upper dials and $a + by$ will appear on the lower dials. In other words, we have now set the machine so that it will readily give $a + by$ as a function of $53{,}518 + y$. Now take $53{,}518 + y = 53{,}550$ and compute $a + by$. Then we see by the list of primes that

$$p(53{,}518 + y + 50 - 1) < a + by,$$

so that

$$53{,}518 + y + 50 < \pi(a + by) + 1,$$

and so (8) holds for $53{,}550 \leq 53{,}518 + y \leq 53{,}600$. Put $53{,}518 + y = 53{,}600$, and then

$$p(53{,}518 + y + 50 - 1) < a + by,$$

so that (8) holds for $53{,}600 \leq 53{,}518 + y \leq 53{,}650$. And so on. In general, for x in the neighborhood of $e^{13.4}$, one can advance y by 50 at a time. Of course irregularities in the distribution of the primes cause trouble occasionally. For instance, if $53{,}518 + y = 53{,}800$ then $\pi(a + by) + 1 = 53{,}848$, so it would appear that one could only advance y by 48. However

This content downloaded from
154.59.125.115 on Sun, 01 Nov 2026 12:04:59 UTC
All use subject to https://about.jstor.org/terms

$p(53{,}850-1)$ only exceeds $a+by$ by eleven, so that one can readily see that if one increased y by 48, then $a+by$ would be enough larger to justify advancing y by 2 more.

For x in other neighborhoods, one would advance y uniformly by some amount different from 50. Three factors determine the choice of this amount:

I. It should be small enough so that exceptional cases occur infrequently.

II. For speed in operating the machine, it should be a multiple of 10 if possible.

III. For convenience in locating entries in the list of primes, it should be a factor or multiple of 100.

THEOREM 2. *For* $0 < x \leq 1{,}000{,}000$, $\theta(x) < x$.

Proof. Clearly it suffices to prove $\theta(p(n)) < p(n)$ for $p(n) \leq 1{,}000{,}000$. As in the proof of Theorem 1, we divide the interval $0 < x \leq 1{,}000{,}000$ into subintervals. So suppose we wish to prove $\theta(p(n+h)) < p(n+h)$ for $n \leq n+h \leq N$. If $p(n) \leq 10{,}000$, compute $\theta(p(n))$. If $p(n) > 10{,}000$, use Theorem 1 and Lemma 5 to compute a k such that $\theta(p(n)) < k$. Then

$$\theta(p(n+h)) < k + h \log N.$$

So it suffices to prove that

$$k + h \log N < p(n+h).$$

That is, it suffices to prove that

$$\pi(k + h \log N) + 1 < n + 1 + h.$$

Put $x = k + h \log N$, and we have the problem reduced to that of comparing $\pi(x)+1$ with $Ax + B + 1$ in a given range, and appropriate modifications of the technique described in the proof of Theorem 1 will work.

COROLLARY. *For* $1 < x \leq 1{,}000{,}000$, $\theta(x) < \left(1 + \dfrac{1}{\log x}\right) x$.

THEOREM 3. *For* $2 \leq x \leq 1{,}000{,}000$, $\pi(x) < li(x)$.

Proof. For $2 \leq x \leq e^{2.4}$, compare values of $\pi(x)$ and $li(x)$. For $e^{2.4} \leq x \leq 1{,}000{,}000$ use Theorem 2 and Lemma 4.

THEOREM 4. *For* $7 \leq x \leq 1{,}000{,}000$, $x - \log x\, li(x^{\frac{1}{2}}) < \theta(x)$.

Proof. For $7 \leq x \leq e^3$, compare values. For $e^3 \leq x \leq 1{,}000{,}000$, use Lemma 6, Cor.

This content downloaded from
154.59.125.115 on Sun, 01 Nov 2026 12:04:59 UTC
All use subject to https://about.jstor.org/terms

Remarks. If $10{,}000 < x \leq 1{,}000{,}000$, and one wishes more exact bounds for $\theta(x)$ than those given in Theorem 2 and Theorem 4, they can be obtained from the List of Prime Numbers with the aid of Lemma 5 and Lemma 6.

It seems likely that Theorems 1-4 remain true if the upper limit of 1,000,000 is replaced by an upper limit of 10,000,000. However it is known that all four theorems are false if one tries to replace the upper limit by infinity (4, Chapter V, especially Theorem 34 and Theorem 35).

THEOREM 5. *For* $0 < x \leq 1420$ *and for* $1423 \leq x \leq 10{,}000$,

$$x - 2x^{\frac{1}{2}} < \theta(x).$$

Proof. Compare values.

Remark. For $x = 1421$ and $x = 1422$, $\theta(x) < x - 2x^{\frac{1}{2}}$.

THEOREM 6. *For* $0 < x \leq 10{,}000$, $x - 2.025807x^{\frac{1}{2}} < \theta(x)$.

THEOREM 7. *For* $0 < x \leq 1{,}000{,}000$, $x - 2.78x^{\frac{1}{2}} < \theta(x)$.

Proof. By Theorem 4 and Theorem 6, one only needs to prove that $\log x \, li(x^{\frac{1}{2}}) \leq 2.78x^{\frac{1}{2}}$ for $10{,}000 \leq x$, and this is not difficult.

A slight additional computation allows us to infer:

COROLLARY. *For* $41 \leq x \leq 1{,}000{,}000$, $\left(1 - \dfrac{1}{\log x}\right)x < \theta(x)$.

THEOREM 8. *For* $17 \leq x \leq 1{,}000{,}000$, $\dfrac{x}{\log x} < \pi(x)$.

Proof. For $17 \leq x \leq e^4$, compare values. For $e^4 \leq x \leq 1{,}000{,}000$, use Theorem 1 and Lemma 8.

THEOREM 9. *For* $e^2 < x \leq 1{,}000{,}000$, $\pi(x) < \dfrac{x}{\log x - 2}$.

Similar proof using Theorem 3 and Lemma 7.

THEOREM 10. *For* $1 < x \leq 83{,}498$, $n \log n + n \log\log n - 2n < p(n)$.

Proof. For $1 < n \leq 1480$, $\log\log n < 2$, so that the theorem follows from the fact that $p(n) > n \log n$ (5, Theorem 1, p. 37). For $1480 \leq n \leq 83{,}498$, $\theta(p(n)) < p(n)$ by Theorem 2 and $li(n) < n$, so that the result follows by Lemma 9.

THEOREM 11. *If* $6 \leq n \leq 83{,}498$, $p(n) < n \log n + n \log\log n$.

Proof. For $6 \leq n \leq e^5$, compare values. For $e^5 \leq n \leq 83{,}498$, we use induction on n. Assume $p(r) < r \log r + r \log\log r$ for $6 \leq r < n$. Then

This content downloaded from
154.59.125.115 on Sun, 01 Nov 2026 12:04:59 UTC
All use subject to https://about.jstor.org/terms

we wish to prove that $p(n) < n \log n + n \log \log n$. So suppose $p(n) \geq n \log n + n \log \log n$. Then

$$\theta(n \log n + n \log \log n) \leq \theta(p(n)).$$

So by Theorem 7 and Lemma 10,

$$n \log n + n \log \log n - 2.78(n \log n + n \log \log n)^{\frac{1}{2}} < n \log n + n \log \log n - n + \frac{n \log \log n}{\log n}.$$

That is,

$$n < 2.78(n \log n + n \log \log n)^{\frac{1}{2}} + \frac{n \log \log n}{\log n},$$

which is false if $n \geq e^5$.

Bounds for $\psi(x)$. Define

$$\phi(x) = \psi(x) - x + \log 2\pi + \tfrac{1}{2} \log(1 - 1/x^2).$$

$$K_m(x, h) = \int_0^h dy_1 \int_0^h dy_2 \cdots \int_0^h \phi(x + y_1 + y_2 + \cdots + y_m)\, dy_m.$$

$$f_{m,n,a}(x, h, z) = \frac{K_m(x, h)}{h^n} + \tfrac{1}{2} n h^a - z h^{a-1}.$$

In the next five lemmas we assume $x > 1$ and $x + mh > 1$ so that the functions and integrals discussed will all exist and take only real values.

LEMMA 11. *Obviously*

$$\int_0^h f_{m,1,a}(x, h, z)\, dz = K_m(x, h).$$

LEMMA 12. *Obviously*

$$\int_0^h f_{m,n,a}(x, h, y_1 + y_2 + \cdots + y_n)\, dy_n = f_{m,n-1,a+1}(x, h, y_1 + y_2 + \cdots + y_{n-1}).$$

LEMMA 13. $K_m(x, h)$

$$= \int_0^h dy_1 \int_0^h dy_2 \cdots \int_0^h f_{m,n,a}(x, h, y_1 + y_2 + \cdots + y_n)\, dy_n.$$

Proof by induction on n, using Lemmas 11-12.

Define

$$f_m(x, h, z) = f_{m,m,1}(x, h, z).$$

LEMMA 14. *If $0 < h$, then there is a z such that $0 \leq z \leq mh$ and*

$$\phi(x + z) \leq f_m(x, h, z).$$

Proof. Suppose that $\phi(x + z) > f_m(x, h, z)$ for $0 \leq z \leq mh$. Then

This content downloaded from 154.59.125.115 on Sun, 01 Nov 2026 12:04:59 UTC
All use subject to https://about.jstor.org/terms

$$\int_0^h dy_1 \int_0^h dy_2 \cdots \int_0^h \phi(x + y_1 + y_2 + \cdots + y_m)\, dy_m$$
$$> \int_0^h dy_1 \int_0^h dy_2 \cdots \int_0^h f_m(x, h, y_1 + y_2 + \cdots + y_m)\, dy_m.$$

However the former equals $K_m(x, h)$ by definition, and the latter equals $K_m(x, h)$ by Lemma 13.

LEMMA 15. *If $h < 0$, then there is a z such that $mh \leq z \leq 0$ and*

$$\phi(x + z) \geq f_m(x, h, z).$$

Proof. Suppose that $\phi(x + z) < f_m(x, h, z)$ for $mh \leq z \leq 0$. Then

$$\int_h^0 dy_1 \int_h^0 dy_2 \cdots \int_h^0 \phi(x + y_1 + y_2 + \cdots + y_m)\, dy_m$$
$$< \int_h^0 dy_1 \int_h^0 dy_2 \cdots \int_h^0 f_m(x, h, y_1 + y_2 + \cdots + y_m)\, dy_m.$$

However the former equals $(-1)^m K_m(x, h)$ by definition, and the latter equals $(-1)^m K_m(x, h)$ by Lemma 13.

THEOREM 12. *If $0 < \delta < (x - 1)/xm$, and*

$$\epsilon_1 = \frac{K_m(x, -x\delta)}{(-x)^{m+1}\delta^m} + \frac{m\delta}{2}, \qquad \epsilon_2 = \frac{K_m(x, x\delta)}{x^{m+1}\delta^m} + \frac{m\delta}{2},$$

then

$$x(1 - \epsilon_1) - \log 2\pi - \tfrac{1}{2}\log(1 - 1/x^2) \leq \psi(x)$$
$$\leq x(1 + \epsilon_2) - \log 2\pi - \tfrac{1}{2}\log(1 - 1/x^2).$$

Proof. Put $h = \delta x$. Use Lemma 14, and put in the definitions of $\phi(x)$ and $f_m(x, h, z)$. Hence there is a z such that $0 \leq z \leq mh$, and

$$\psi(x + z) - (x + z) + \log 2\pi + \tfrac{1}{2}\log\left(1 - \frac{1}{(x + z)^2}\right) \leq \frac{K_m(x, h)}{h^m} + \frac{mh}{2} - z.$$

Replace h by $x\delta$, and one has

$$\psi(x + z) \leq x(1 + \epsilon_2) - \log 2\pi - \tfrac{1}{2}\log\left(1 - \frac{1}{(x + z)^2}\right).$$

However, $0 \leq z$, so that $\psi(x) \leq \psi(x + z)$ and

$$-\tfrac{1}{2}\log\left(1 - \frac{1}{(x + z)^2}\right) \leq -\tfrac{1}{2}\log(1 - 1/x^2).$$

This proves half the theorem. To prove the other half, put $h = -x\delta$, and proceed similarly, using Lemma 15.

Henceforth, we denote non-trivial zeros (4, p. 58) of $\zeta(s)$ by $\rho = \beta + i\gamma$.

This content downloaded from
154.59.125.115 on Sun, 01 Nov 2026 12:04:59 UTC
All use subject to https://about.jstor.org/terms

THEOREM 13. $\int_0^h \phi(x+z)\,dz = \sum_\rho \frac{1}{\rho(\rho+1)}\{x^{\rho+1} - (x+h)^{\rho+1}\}.$

Proof. It is known (6, p. 317) that

$$\frac{\zeta'(0)}{\zeta(0)} = \log 2\pi,$$

and also (4, p. 30, p. 73) that

$$\int_1^x \psi(u)\,du = \frac{x^2}{2} - \sum_\rho \frac{x^{\rho+1}}{\rho(\rho+1)} - x\frac{\zeta'(0)}{\zeta(0)} + \frac{\zeta'(-1)}{\zeta(-1)} - \sum_{r=1}^{\infty} \frac{x^{1-2r}}{2r(2r-1)}.$$

The theorem follows from these and the fact that $\sum_\rho |\rho|^{-2}$ is convergent (4, Theorem 18, p. 57). This latter fact enables us to integrate the result of Theorem 13 term by term and deduce Theorem 14.

THEOREM 14. $K_m(x, \pm x\delta) =$

$$\sum_\rho \frac{x^{\rho+m}}{\rho(\rho+1)\cdots(\rho+m)}\{\sum_{j=0}^{m} (-1)^{j+m+1}\binom{m}{j}(1 \pm j\delta)^{\rho+m}\}.$$

Define

$$K = K(m, x) = \sum_\rho \frac{x^{\beta-1}}{|\gamma^{m+1}|}. \tag{9}$$

THEOREM 15. *If* $0 \leq \delta$, *then*

$$|K_m(x, \pm x\delta)| < x^{m+1}((1+\delta)^{m+1} + 1)^m K.$$

Proof. Take absolute values of both sides of Theorem 14.

$$\left|\frac{x^{\rho+m}}{\rho(\rho+1)\cdots(\rho+m)}\right| < \frac{x^{\beta+m}}{|\gamma^{m+1}|}$$

(4, Theorem 16, p. 48). Also

$$|(1 \pm j\delta)^{\rho+m}| = (1 \pm j\delta)^{\beta+m} < (1+j\delta)^{m+1}.$$

So

$$\begin{aligned}
|\sum_{j=0}^{m} (-1)^{j+m+1}\binom{m}{j}(1 \pm j\delta)^{\rho+m}| &< \sum_{j=0}^{m}\binom{m}{j}(1+j\delta)^{m+1} \\
&< \sum_{j=0}^{m}\binom{m}{j}((1+\delta)^j)^{m+1} \\
&= \sum_{j=0}^{m}\binom{m}{j}((1+\delta)^{m+1})^j \\
&= ((1+\delta)^{m+1} + 1)^m.
\end{aligned}$$

THEOREM 16. *With* K *as in* (9) *and* ϵ_1 *and* ϵ_2 *as in Theorem* 12, *if one takes*

This content downloaded from 154.59.125.115 on Sun, 01 Nov 2026 12:04:59 UTC
All use subject to https://about.jstor.org/terms

$$\delta \geq 2K^{1/(m+1)}, \qquad \theta = \frac{\delta}{2}\left\{\left(\frac{(1+\delta)^{m+1}+1}{2}\right)^{m} + m\right\},$$

then $\epsilon_1 < \theta$ and $\epsilon_2 < \theta$.

Proof by use of Theorem 15.

We now derive an upper bound for K. First we need information about the zeros of $\zeta(s)$. Let $N(T)$ denote the number of ρ's for which $0 < \gamma \leq T$. Define

$$F(T) = \frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi} + \frac{7}{8},$$

$$R(T) = 0.137 \log T + 0.443 \log\log T + 1.588.$$

THEOREM 17. *For* $0 \leq T \leq 280$, $|N(T) - F(T)| < 1$.

Proof. This can be deduced from the computations of Hutchinson (7, pp. 49-60).

Choose A so that $F(A) = 1041$. Then $A = 1467.47747$ correct to five decimal places.

THEOREM 18. *For* $0 < T \leq A$, $|N(T) - F(T)| < 2$; $N(A) = F(A) = 1041$; *and for* $0 < \gamma \leq A$, $\beta = \frac{1}{2}$.

Proof. The computations of Titchmarsh (8, pp. 234-250, and 9, pp. 261-263) are almost adequate to give these results, the missing computation being that which proves that $N(A) < 1043$. This computation has been performed by the present author, using a variation of a method of Titchmarsh (8, pp. 251-252).

THEOREM 19. *For* $2 \leq T$, $|N(T) - F(T)| < R(T)$.

Proof. For $2 \leq T \leq A$, the theorem follows from Theorems 17-18. For $A \leq T$, apply the method of Backlund (10, pp. 354-375) to $(\zeta(s))^N$ instead of to $\zeta(s)$. For instance, the first part of Backlund's discussion (10, pp. 354-361) when applied to $(\zeta(s))^N$ yields

$$|N(T) - F(T)| < \frac{1}{2\pi qN}\int_0^{\pi} \log|G(re^{i\phi})|\, d\phi + \frac{1}{2} + \frac{4}{\pi T} + \frac{k}{N} - \frac{1}{2qN}\log|G(0)| + \frac{1}{\pi}|S(T)|,$$

where k is a constant independent of N, $r = 1.32$,

This content downloaded from 154.59.125.115 on Sun, 01 Nov 2026 12:04:59 UTC
All use subject to https://about.jstor.org/terms

$$q = \log \frac{4r}{3} = 0.565314,$$

$$G(s) = \frac{1}{2}\{(\zeta(s + \frac{5}{4} + Ti))^N + (\zeta(s + \frac{5}{4} - Ti))^N\},$$

$$S(T) = \Im \log \Gamma\left(\frac{1}{4} + \frac{Ti}{2}\right) - \frac{T}{2}\log\frac{T}{2} + \frac{T}{2} + \frac{\pi}{8}.$$

From Backlund's bounds for $|\zeta(s)|$ (10, pp. 361-369) and the fact that $A \leq T$, one can deduce upper bounds for $|G(re^{i\phi})|$. To deduce an upper bound for $-\log|G(0)|$, note that

$$G(0) = \Re(\zeta(5/4 + iT)^N).$$

Hence we can choose a succession of N's tending to infinity so that

$$\operatorname{Lim} \frac{G(0)}{|\zeta(5/4 + iT)^N|} = 1.$$

However (5, p. 26),

$$|\zeta(5/4 + iT)| \geq \frac{1}{3.425}.$$

All other steps in the proof are strictly analogous to the steps of Backlund's proof.

THEOREM 20. *For* $A \leq \gamma$, $\beta < 1 - \dfrac{1}{17.72 \log \gamma}$.

Proof. Use the procedure outlined by Landau (6, pp. 318-324) using the numerical evaluations of Rosser (5, pp. 30-31) and the fact that

$$18 + 30\cos\phi + 17\cos 2\phi + 6\cos 3\phi + \cos 4\phi = 2(1 + \cos\phi)^2(1 + 2\cos\phi)^2 \geq 0.$$

Having no further use for the old meaning of $\phi(x)$, we now define

$$\phi(\gamma) = \phi(m, x, \gamma) = \frac{x^{-1/(17.72 \log \gamma)}}{\gamma^{m+1}}.$$

LEMMA 16. *With* K *as in* (9)

$$K \leq x^{-\frac{1}{2}} \sum_{\rho} \frac{1}{|\gamma^{m+1}|} + \sum_{A < \gamma} \phi(\gamma).$$

Proof. By definition,

$$K = \sum_{\beta \leq \frac{1}{2}} \frac{x^{\beta-1}}{|\gamma^{m+1}|} + \sum_{\beta > \frac{1}{2}} \frac{x^{\beta-1}}{|\gamma^{m+1}|}.$$

However

$$\sum_{\beta \leq \frac{1}{2}} \frac{x^{\beta-1}}{|\gamma^{m+1}|} \leq x^{-\frac{1}{2}} \sum_{\beta \leq \frac{1}{2}} \frac{1}{|\gamma^{m+1}|} \leq x^{-\frac{1}{2}} \sum_{\rho} \frac{1}{|\gamma^{m+1}|}.$$

This content downloaded from 154.59.125.115 on Sun, 01 Nov 2026 12:04:59 UTC
All use subject to https://about.jstor.org/terms

Since $\beta - i\gamma$ is a zero of $\zeta(s)$ if $\beta + i\gamma$ is (4, Theorem 16, p. 48),

$$\sum_{\beta>\frac{1}{2}} \frac{x^{\beta-1}}{|\gamma^{m+1}|} = 2 \sum_{\substack{\beta>\frac{1}{2} \\ \gamma>0}} \frac{x^{\beta-1}}{\gamma^{m+1}}.$$

However, by Theorem 20,

$$\frac{x^{\beta-1}}{\gamma^{m+1}} < \phi(\gamma).$$

So

$$\sum_{\beta>\frac{1}{2}} \frac{x^{\beta-1}}{|\gamma^{m+1}|} < 2 \sum_{\substack{\beta>\frac{1}{2} \\ \gamma>0}} \phi(\gamma).$$

By Theorem 18, if $\beta > \frac{1}{2}$ and $\gamma > 0$, then $\gamma > A$. Also $1 - \beta + i\gamma$ is a zero of $\zeta(s)$ if $\beta + i\gamma$ is (4, Theorem 18, p. 48). So

$$2 \sum_{\substack{\beta>\frac{1}{2} \\ \gamma>0}} \phi(\gamma) \leq \sum_{\gamma>A} \phi(\gamma).$$

LEMMA 17. $\sum_{\rho} \frac{1}{\gamma^2} < 0.0463$; $\sum_{\rho} \frac{1}{|\gamma^3|} < 0.00167$; $\sum_{\rho} \frac{1}{\gamma^4} < 0.0000744$.

Proof. The first result has been proved before (5, pp. 28-30). From it we deduce the second result as follows. Compute

$$\sum_{0<\gamma\leq 50} \frac{1}{\gamma^2}$$

from Gram's values of γ (11, p. 297). Hence deduce an upper bound for

$$\sum_{50<\gamma} \frac{1}{\gamma^2}.$$

Dividing by 52.970 (7, p. 59) gives an upper bound for

$$\sum_{50<\gamma} \frac{1}{\gamma^3}$$

The second result now follows readily. For the third result, note that Gram (11, p. 293) gives the result

$$\tfrac{1}{4} \sum_{\rho} \frac{1}{\alpha^4} = .0000\ 1858\ 6299\ 6426 \cdots,$$

where $\alpha = \gamma + (\beta - \frac{1}{2})i$. Remembering that $\beta = \frac{1}{2}$ if $0 < \gamma \leq A$, we readily deduce the third result.

LEMMA 18. *If* $\log x \leq 942(m+1)$, then

$$\sum_{A<\gamma} \phi(\gamma) < \frac{3.47}{A^{m+1}x^{1/130}} + 0.1592 \int_A^{\infty} \log \frac{y}{2\pi}\, \phi(y)\, dy.$$

This content downloaded from
154.59.125.115 on Sun, 01 Nov 2026 12:04:59 UTC
All use subject to https://about.jstor.org/terms

Proof. By a theorem connecting sums with integrals (4, Theorem A, p. 18),

$$\sum_{A<\gamma}\phi(\gamma) = -\int_A^\infty N(y)\phi'(y)\,dy - N(A)\phi(A).$$

However for $A \leq y$,

$$\log x \leq 942(m+1) < 17.72(m+1)\log^2 A \leq 17.72(m+1)\log^2 y,$$

and so $\phi'(y) < 0$. Hence by Theorem 19

$$\sum_{A<\gamma}\phi(\gamma) < -\int_A^\infty (F(y)+R(y))\phi'(y)\,dy - N(A)\phi(A).$$

Integrating by parts gives

$$\sum_{A<\gamma}\phi(\gamma) < \int_A^\infty (F'(y)+R'(y))\phi(y)\,dy + R(A)\phi(A).$$

However

$$R(A)\phi(A) < \frac{3.47}{A^{m+1}x^{1/130}},$$

$$\int_A^\infty F'(y)\phi(y)\,dy = \frac{1}{2\pi}\int_A^\infty \log\frac{y}{2\pi}\,\phi(y)\,dy,$$

and

$$\int_A^\infty R'(y)\phi(y)\,dy = 0.137\int_A^\infty \frac{\phi(y)\,dy}{y} + 0.443\int_A^\infty \frac{\phi(y)\,dy}{y\log y}$$
$$< \frac{0.137}{A\log\frac{A}{2\pi}}\int_A^\infty \log\frac{y}{2\pi}\,\phi(y)\,dy + \frac{0.443}{A\log A\log\frac{A}{2\pi}}\int_A^\infty \log\frac{y}{2\pi}\,\phi(y)\,dy.$$

LEMMA 19. *If*

$$\log x < \frac{942m^2}{m+0.184},$$

then

$$\int_A^\infty \log\frac{y}{2\pi}\,\phi(y)\,dy < \frac{1+5.454m}{\left(1-\frac{m+0.184}{942m^2}\log x\right)m^2A^mx^{1/130}}.$$

Proof. Integrating the left hand side by parts, using

$$\frac{\log\frac{y}{2\pi}}{y^{m+1}}$$

as the part to be integrated, we get

$$\int_A^\infty \log\frac{y}{2\pi}\,\phi(y)\,dy = \frac{1+m\log\frac{A}{2\pi}}{m^2A^mx^{1/17.72\log A}}$$
$$+\int_A^\infty \frac{\left(m+1/\log\frac{y}{2\pi}\right)\log x}{17.72m^2\log^2 y}\log\frac{y}{2\pi}\,\phi(y)\,dy.$$

So

This content downloaded from
154.59.125.115 on Sun, 01 Nov 2026 12:04:59 UTC
All use subject to https://about.jstor.org/terms

$$\int_A^\infty \log\frac{y}{2\pi}\,\phi(y)\,dy < \frac{1+5.454m}{m^2A^mx^{1/130}} + \frac{m+1/\log\dfrac{A}{2\pi}\log x}{17.72m^2\log^2 A}\int_A^\infty \log\frac{y}{2\pi}\,\phi(y)\,dy.$$

Solving for $\int_A^\infty \log\frac{y}{2\pi}\,\phi(y)\,dy$ proves the theorem.

THEOREM 21. If

$$\sum_\rho \frac{1}{|\gamma^{m+1}|} \leq k, \qquad \log a < \frac{942m^2}{m+0.184},$$

$$\delta = 2\left\{a^{-\frac{1}{2}}k + \frac{3.47}{A^{m+1}a^{1/130}} + \frac{0.869m+0.160}{(1-\dfrac{m+0.184}{942m^2}\log a)\,m^2A^ma^{1/130}}\right\}^{1/(m+1)},$$

$$\epsilon = \frac{\delta}{2}\left\{\left(\frac{(1+\delta)^{m+1}+1}{2}\right)^m + m\right\},$$

and $1 + m\epsilon a < a$, then for $a \leq x$,

$$x(1-\epsilon) - 1.84 < \psi(x) < x(1+\epsilon) - \tfrac{1}{2}\log(1-1/x^2).$$

Proof. Since $K(m,x) \leq K(m,a)$ if $a \leq x$, the theorem follows from Theorem 12, Theorem 16, Lemma 16, Lemma 18, and Lemma 19.

By use of this theorem, Table II was computed. In those cases where $m = 4$ or $m = 5$, x was large enough so that one could use 0.0000744 (see Lemma 17) as an upper bound for

$$\sum_\rho \frac{1}{|\gamma^5|} \quad \text{and} \quad \sum_\rho \frac{1}{\gamma^6}$$

without appreciably affecting the value of ϵ.

By a method of Rosser (5, pp. 39-40 and Lemma 10, p. 31) one can prove Theorem 22.

THEOREM 22. *If* $\epsilon(x) = (\log x)^{\frac{1}{2}}e^{-\sqrt{(\log x/19)}}$ *and* $e^{4000} \leq x$, *then*

$$(1-\epsilon(x))x < \psi(x) < (1+\epsilon(x))x.$$

Remarks. From (2), Theorem 2, Theorem 7, Table II, Theorem 22, and Gram's table of $\psi(x)$, one can deduce:

For $1 < x \leq e^{100}$ and for $e^{2000} \leq x$,

$$\left(1-\frac{1}{\log x}\right)x < \psi(x) < \left(1+\frac{1}{\log x}\right)x.$$

For $1 < x$,

$$\left(1-\frac{2.85}{\log x}\right)x < \psi(x) < \left(1+\frac{2.85}{\log x}\right)x.$$

This content downloaded from 154.59.125.115 on Sun, 01 Nov 2026 12:04:59 UTC
All use subject to https://about.jstor.org/terms

A curious fact is that $\psi(x)/x$ takes its maximum at $x = 113$, so that one can say that

$$\psi(x) < 1.038821x$$

for all positive x.

In an earlier draft of this paper, a much more complicated, but more accurate, bound for K (depending on an improvement of Theorem 20) was used to deduce the result:

For $1 < x$,

$$\left(1 - \frac{1.8}{\log x}\right)x < \psi(x) < \left(1 + \frac{1.8}{\log x}\right)x.$$

However the computations supporting this result were quite extensive and have never been checked. It is the author's hope to eventually improve the upper bound for K so that one can prove:

For $e^{80} < x$,

$$\left(1 - \frac{0.95}{\log x}\right)x < \psi(x) < \left(1 + \frac{0.95}{\log x}\right)x.$$

If this could be done, one could replace the upper bounds on x and n in Theorems 23-28 by infinity.

The range $1{,}000{,}000 \leq x$. By (2) and Table II we may infer that:

(10) For $e^{13.8} \leq x$, $(1 - 0.0393)x < \theta(x) < (1 + 0.0376)x$.

(11) For $e^{15} \leq x$, $(1 - 0.0328)x < \theta(x) < (1 + 0.0321)x$.

(12) For $e^{20} \leq x$, $(1 - 0.02)x < \theta(x) < (1 + 0.0199)x$.

For $e^{30} \leq x$, we may take Table II and Theorem 22 as referring to $\theta(x)$ as well as to $\psi(x)$. Hence, with the help of Theorem 2, Corollary and Theorem 7, Corollary one can readily deduce Theorems 23, 24, 29C, and 30C. Similarly, one can deduce Lemmas 20-21.

LEMMA 20. *For* $2 \leq x$,

$$\left(1 - \frac{2.85}{\log x}\right)x < \theta(x) < \left(1 + \frac{2.85}{\log x}\right)x.$$

LEMMA 21. *For* $e^{2000} \leq x$,

$$\left(1 - \frac{0.96}{\log x}\right)x < \theta(x) < \left(1 + \frac{0.96}{\log x}\right)x.$$

Remarks. Because of Theorem 2 and (10), we may infer that $\theta(x) < 1.0376x$ for all positive x. Hence

$$\prod_{p \leq x} p < (2.83)^x$$

This content downloaded from
154.59.125.115 on Sun, 01 Nov 2026 12:04:59 UTC
All use subject to https://about.jstor.org/terms

for positive x. Because of Theorem 5, Theorem 7, and (10), we may infer that $0.9607x < \theta(x)$ for $2600 \leq x$. Computation and Theorem 6 gives the result that $0.6932x < \theta(x)$ for $29 \leq x$. Hence

$$(2.61)^x < \prod_{p \leq x} p \text{ for } 2600 \leq x.$$

$$2^x < \prod_{p \leq x} p \text{ for } 29 \leq x.$$

In order to derive Theorems 25, 26, 29A, and 30A, we proceed as follows. First note that by (7) and Theorem 24,

$$\pi(x) < \frac{x}{\log x} + \frac{x}{\log^2 x} + \int_2^x \frac{dy}{\log^2 y} + \int_2^x \frac{dy}{\log^3 y} \tag{13}$$

for $2 \leq x \leq e^{100}$. Since

$$\int \frac{dy}{\log^3 y} = -\frac{y}{2\log^2 y} + \int \frac{dy}{2\log^2 y}$$

and

$$\int \frac{dy}{\log^2 y} = -\frac{y}{\log y} + li(y),$$

one can compute the value of the right side of (13) for any x. In particular one can show that for $e^{13.8} \leq x$, the right side of (13) is less than $x/(\log x - 2)$. Theorem 25 now follows by use of Theorem 9.

By (7),

$$\pi(x) > \frac{\theta(x)}{\log x} + \int_{41}^x \frac{\theta(y)\,dy}{y \log^2 y}$$

if $41 \leq x$. So for $41 \leq x \leq e^{100}$,

$$\pi(x) > \frac{x}{\log x} - \frac{x}{\log^2 x} + \int_{41}^x \frac{dy}{\log^2 y} - \int_{41}^x \frac{dy}{\log^3 y}$$

by Theorem 23. For $e^{13.8} \leq x$, the right side of this is greater than $x/\log x$. So Theorem 26 follows by use of Theorem 8.

Now assume $e^{100} \leq x$. Then by (7) and Lemma 20

$$\pi(x) < \frac{x}{\log x} + \frac{2.85x}{\log^2 x} + \int_2^x \frac{dy}{\log^2 y} + 2.85 \int_2^x \frac{dy}{\log^3 y}. \tag{14}$$

However

$$\int_2^x \frac{dy}{\log^2 y} = \frac{x}{\log^2 x} - \frac{2}{\log^2 2} + 2\int_2^x \frac{dy}{\log^3 y} < \frac{x}{\log^2 x} + 2\int_2^x \frac{dy}{\log^3 y},$$

$$\int_2^x \frac{dy}{\log^3 y} < \frac{x}{\log^3 x} + 3\int_2^x \frac{dy}{\log^4 y},$$

and

$$\int_2^x \frac{dy}{\log^4 y} = \int_2^{x^{\frac{1}{2}}} \frac{dy}{\log^4 y} + \int_{x^{\frac{1}{2}}}^x \frac{dy}{\log^4 y} < x^{\frac{1}{2}} + \frac{x}{\log^4 (x^{\frac{1}{2}})}.$$

This content downloaded from
154.59.125.115 on Sun, 01 Nov 2026 12:04:59 UTC
All use subject to https://about.jstor.org/terms

From these inequalities, one can easily show that the right side of (14) is less than $x/(\log x - 4)$, and so prove one half of Theorem 29A. The proof of the other half is similar.

Now assume $e^{2000} \leq x$. Then by (7) and Lemmas 20-21,

$$\pi(x) < \frac{x}{\log x} + \frac{0.96x}{\log^2 x} + \int_2^x \frac{dy}{\log^2 y} + 2.85 \int_2^x \frac{dy}{\log^3 y}.$$

This is treated in the same manner that (14) was, and one half of Theorem 30A is proved thereby. The other half is proved in a similar manner.

We now consider Theorem 27, Theorem 29B, and Theorem 30B. To prove the last two, we will need to prove the results

$$(n+1)\log(n+1)+(n+1)\log\log(n+1)-(n+1)-A(n+1)<p(n)$$

for $A = 3$ and $A = 1$ respectively. Let $a \leq p(n)$, and suppose that for $a \leq x$, $\theta(x) < (1+\epsilon)x$. Let $80{,}000 \leq n$. Now suppose that

$$p(n) \leq (n+1)\log(n+1) + (n+1)\log\log(n+1) - 2(n+1).$$

Since $80{,}000 \leq n$, we can infer

$$p(n) < n\log n + n\log\log n - 2n + 2\log n.$$

Then by Lemma 9,

$$n\log n + n\log\log n - n - li(n) < (1+\epsilon)(n\log n + n\log\log n - 2n + 2\log n).$$

That is

$$1 + 2\epsilon < \frac{li(n)}{n} + \frac{(2+2\epsilon)\log n}{n} + \epsilon(\log n + \log\log n). \tag{15}$$

Now if $a = e^{13.8}$ and $a \leq p(n)$, then $82{,}395 \leq n$. Also, by (10), we can take $\epsilon = 0.0376$. Then (15) is false for $n \leq e^{20}$. If $a = e^{20}$, we can take $\epsilon = 0.0199$ by (12). Then (15) is false for $n \leq e^{40}$. If $a = e^{40}$, we can take $\epsilon = 0.0119$ by Table II. Then (15) is false for $n \leq e^{80}$. And so on up to $n \leq e^{100}$. This with Theorem 10 proves Theorem 27, and a similar procedure proves one half of Theorem 29B and Theorem 30B.

We now consider Theorem 28. Suppose $a \leq p(n)$, and that for $a \leq x$, $(1-\epsilon)x < \theta(x)$. Also suppose that $p(r) < r\log r + r\log\log r$ for $16 \leq r < n$ and that $n\log n + n\log\log n \leq p(n)$. Then by Lemma 10,

This content downloaded from 154.59.125.115 on Sun, 01 Nov 2026 12:04:59 UTC
All use subject to https://about.jstor.org/terms

$$(1-\epsilon)(n\log n + n\log\log n) < n\log n + n\log\log n - n + \frac{n\log\log n}{\log n}.$$

That is,

$$1 < \frac{\log\log n}{\log n} + \epsilon(\log n + \log\log n).$$

For $83{,}000 \leq n \leq e^{95}$, this is false, and so Theorem 28 follows.

LEMMA 22. *If* $e^{95} \leq n$, *then*

$$\theta(p(n)) < n\log n + n\log\log n - n + \frac{2n\log\log n}{\log n}.$$

Proof. Since $p(n) < n\log n + 2n\log\log n$ (5, Theorem 2, p. 40),

$$\theta(p(n)) < \theta(p(15)) + \sum_{r=16}^{n-1}\log(r\log r + 2r\log\log r) + \log(n\log n + 2n\log\log n).$$

From here, proceed as in the proof of Lemma 10.

By use of Lemma 22, we can readily prove the rest of Theorem 29B and Theorem 30B.

TABLE I.

x	$\theta(x)$	x	$\theta(x)$
2000	1939.839 200	6500	6408.907 367
2500	2433.602 748	7000	6920.421 031
3000	2932.359 205	7500	7364.857 418
3500	3409.457 181	8000	7875.150 386
4000	3911.145 393	8500	8343.999 665
4500	4412.188 301	9000	8870.374 997
5000	4911.695 346	9500	9418.368 776
5500	5391.372 236	10000	9895.991 380
6000	5893.297 458		

TABLE II.

Against values of b are tabulated those values of ϵ such that one can deduce from Theorem 21 that $(1-\epsilon)x < \psi(x) < (1+\epsilon)x$ for $e^b \leq x$. Since the value of ϵ that can be deduced from Theorem 21 depends on a preassigned value of m, the values of m which were used are tabulated with b and ϵ.

This content downloaded from 154.59.125.115 on Sun, 01 Nov 2026 12:04:59 UTC
All use subject to https://about.jstor.org/terms

b	m	ϵ	b	m	ϵ
13.8	2	0.0381	500	3	0.00511
15	2	0.0321	550	3	0.00467
20	2	0.0199	600	3	0.00427
30	2	0.0179	650	3	0.00392
40	3	0.0119	700	3	0.00356
60	4	0.0101	800	2	0.00294
80	4	0.00983	900	2	0.00235
90	5	0.00938	1000	2	0.00194
100	5	0.00932	1300	2	0.00104
300	4	0.00710	2000	3	0.000383
400	4	0.00609	2300	3	0.000255
450	4	0.00567	3000	4	0.000158

PRINCETON UNIVERSITY.

BIBLIOGRAPHY.

1. J. P. Gram, "Maengden af Primtal under en given Graense," *K. Danske Vidensk. Selskabs Skrifter*, ser. 6, vol. 2 (1881-86), pp. 183-308.

2. G. W. Jones, *Logarithmic Tables*, Macmillan and Co., Seventh Edition, 1898.

3. D. N. Lehmer, *List of Prime Numbers from* 1 *to* 10,006,721, Carnegie Institution of Washington Publication No. 165, 1914.

4. A. E. Ingham, *The Distribution of Prime Numbers*, Cambridge Tract No. 30, 1932, London.

5. J. B. Rosser, "The n-th prime is greater than $n \log n$," *Proceedings of the London Mathematical Society*, ser. 2, vol. 45 (1939), pp. 21-44.

6. E. Landau, *Handbuch der Lehre von der Verteilung der Primzahlen.*

7. J. I. Hutchinson, "On the roots of the Riemann zeta function," *Transactions of the American Mathematical Society*, vol. 27 (1925), pp. 49-60.

8. E. C. Titchmarsh, "The zeros of the Riemann zeta-function," *Proceedings of the Royal Society of London*, Series A, vol. 151 (1935), pp. 234-255.

9. E. C. Titchmarsh, "The zeros of the Riemann zeta-function," *Proceedings of the Royal Society of London*, Series A, vol. 157 (1936), pp. 261-263.

10. R. J. Backlund, "Über die Nullstellen der Riemannschen Zetafunktion," *Acta Mathematica*, vol. 41 (1917), pp. 345-375.

11. J. P. Gram, "Note Sur les Zéros de la Fonction $\zeta(s)$ de Riemann," *Acta Mathematica*, vol. 27 (1903), pp. 289-304.

This content downloaded from
154.59.125.115 on Sun, 01 Nov 2026 12:04:59 UTC
All use subject to https://about.jstor.org/terms